



101

Workflow automation – for complete beginners.

Learn what n8n is, its core concepts, key features, integration ecosystem and real-world use cases – all in one concise guide.

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WHAT IS n8n?

A workflow automation platform – as easy to start as no-code, as flexible as code when you need it.

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n8n (pronounced "**n-eight-n**", short for *nodemation*) is an open workflow automation platform. You connect apps by dragging and dropping blocks on a visual canvas: when something happens in app A, n8n automatically processes the data and takes action in apps B, C and D – no one needs to press a button.

400+

built-in integrations, plus the HTTP Request node to connect to almost any API

200K+

teams worldwide use it, from startups to enterprises

Fair-code

source-available under a fair-code license – self-host it on your own infrastructure

Why automate?

Copying data between apps, sending reminder emails, compiling reports... these tasks are repetitive, time-consuming and error-prone. Automation frees up your time for work that matters more.

How is n8n different?

When drag-and-drop isn't enough, you can add JavaScript or Python code at any step. And because you can self-host, your data can stay on your own infrastructure – with no per-task pricing.



Think of n8n as a **digital assistant** working 24/7: watching for events, processing data and taking action for you – exactly following the process you design.

7 CONCEPTS TO REMEMBER

Master these 7 concepts and you'll have the foundation to understand how n8n works.

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**Workflow – an automated process**

An automated process built on the canvas. Example: "When a new order email arrives → add it to Google Sheets → notify Slack".

**Node – one step in a workflow**

Each block on the canvas is a node that performs one task: read an email, call an API, transform data, send a message. Workflow = nodes connected together.

**Trigger – where a workflow starts**

A special node that decides *when* the workflow runs: on a schedule (every morning at 8), on an event (a new form submission), or via a webhook from another system.

**Execution – one run of a workflow**

Each time a workflow is triggered is one execution. n8n keeps a history of every run with the data at each step – making it easy to inspect and fix errors.

**Credential – authentication details**

The details n8n uses to connect and authenticate with services (Google accounts, API keys...). Enter once, stored encrypted, reused across workflows.

**Action – nodes that do things**

Nodes placed after the trigger that perform specific operations: send an email, write to a database, post a message to Slack...

**Logic & Control – routing the data flow**

Conditional nodes like IF, Filter, Switch and Merge – used to filter data or branch the execution path.

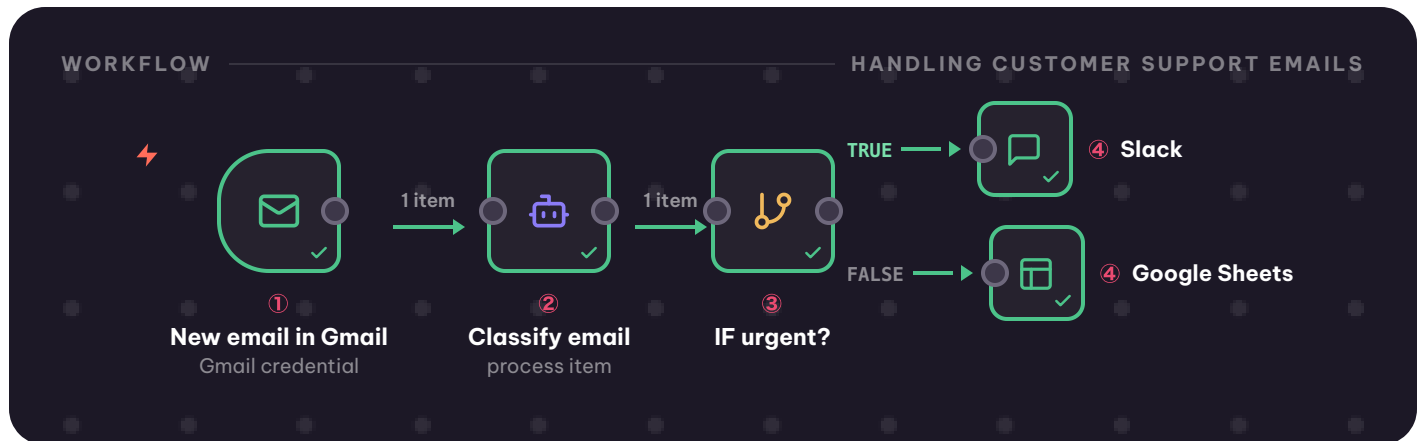
TIP

Data flows through a workflow as **items** (JSON records). Each node receives items from the previous node, processes them, and passes them on to the next.

READING A WORKFLOW

Example: automatically handling customer support emails.

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The whole diagram is a **workflow**; data (*items*) flows from left to right through each **node**. Each run is an **execution**.

① Trigger starts it

A customer emails the support inbox. The trigger detects the new email and starts the workflow – connecting via a saved credential.

② Node processes data

The email becomes an item flowing into an AI node – which analyzes the content and rates its urgency.

③ Branching logic

The IF node checks the "urgent?" condition and routes data to the TRUE or FALSE branch.

④ Action is taken

Urgent branch: the Slack node notifies the team right away. Normal branch: the Google Sheets node logs it for later.

NOTE

You can test each node individually and see the input/output data at every step – so you can check your workflow thoroughly before it runs for real.

KEY FEATURES

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Easy to start, flexible enough for complex workflows.



Visual drag-and-drop editor

Build workflows visually, like drawing a diagram. One look at the canvas shows what the system does – even if you can't code.



Code when you need it

Add JavaScript or Python at any step to handle complex logic – when built-in nodes aren't enough.



Thousands of ready-made templates

A library of community-shared workflows: import, set up credentials and customize to your needs.



Self-host or use Cloud

Run on your own server for full control over your data, or use n8n Cloud without managing any infrastructure.



Error handling & retries

Dedicated error workflows, automatic retries and detailed execution history – to catch and fix errors quickly.



Human in the loop

Add a human approval step to your workflow: send a request via Slack/email, wait for approval, then continue.

AI IN n8n

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Not just automation – you can bring AI into your processes to analyze and make decisions.

n8n lets you build AI-powered workflows right on the platform (built on LangChain): drag an AI node onto the canvas like any other node, choose a model (OpenAI, Anthropic, Google, local models...) and connect it to your data.



AI Agent

An agent takes a request and chooses the right tools to complete it: query a database, call an API, send an email. You define the tools – the agent decides how to use them.



RAG – answers based on your data

Load internal documents into a vector store so AI answers based on the sources you provide. Example: a chatbot that answers company policy questions for new hires.



Chat Trigger

Add a chat interface to your workflow: a built-in chat UI you can embed on a website or connect to Telegram, Slack or WhatsApp.



Flexible AI model choice

Switch models easily – try GPT, Claude, Gemini or a self-hosted model to find the right balance of quality and cost.



A common example: **AI reads a support email → searches the internal knowledge base → drafts a reply → a human reviews → send.** AI helps with the work; people keep control and make the final call.

INTEGRATION ECOSYSTEM

400+ built-in integrations – and connect to any other API via HTTP Request.

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Communication & collaboration

Slack Gmail Telegram Discord
Microsoft Teams WhatsApp

Data & documents

Google Sheets Notion Airtable
PostgreSQL MySQL Google Drive

AI & machine learning

OpenAI Anthropic Google Gemini
Ollama Pinecone Hugging Face

Sales & marketing

HubSpot Salesforce Shopify
Stripe Mailchimp Facebook

Software development

GitHub GitLab Jira AWS
Docker

Can't find the app you need?

HTTP Request and **Webhook** let you connect to almost any service with an API – including your company's internal systems.

The list above shows only highlights – see the full list at n8n.io/integrations. Product names belong to their respective owners.

IDEAS BY DEPARTMENT

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What could you start automating this week?

Marketing

- Auto-publish posts to multiple social networks from one content sheet
- Push leads from forms or ads into your CRM and send a welcome email instantly
- Campaign performance reports sent to the team automatically every Monday morning

Sales & Support

- AI classifies and prioritizes customer emails, flagging urgent cases in the team chat
- Follow-up reminders for deals about to go stale, with automatic CRM status updates
- A chatbot that answers FAQs based on product documentation

Operations & HR

- New hire onboarding: create accounts, add to teams, send a checklist
- Save invoice attachments from email to Google Drive and log the details in a spreadsheet
- Collect leave requests and send them to managers for approval

IT & Data

- Sync data between databases, APIs and internal tools
- Incident alerts with context sent to the on-call operations channel
- Auto-handle repetitive tickets: password resets, access requests

How to choose: the task that repeats daily, follows clear rules and takes the most time – that's usually the best candidate to automate first.

CASE STUDIES

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How are companies using n8n?

Delivery Hero

Food delivery platform, 70+ countries, 53,000+ employees

200 hours

saved every month, from a single workflow

Around 800 employees get locked out of their accounts each month, and each case took IT about 35 minutes. They built one n8n workflow: employees request recovery via Slack, a manager approves, and n8n automatically runs the required steps in Okta and Google Workspace. Lockout time dropped from 35 to 20 minutes – and paved the way for many more IT processes to be automated.

StepStone

Europe's leading online recruitment platform

25× faster

integrating new data sources:
2 weeks → 2 hours

StepStone uses n8n for backend data integration, running 200+ mission-critical workflows every day. Each new data source used to take about 2 weeks of development; with n8n it takes about 2 hours – meeting customization needs while keeping data secure.

Vodafone

Global telecommunications group

~£2.2M

in operating costs saved

Vodafone's cybersecurity team uses n8n to automate threat intelligence flows, orchestrating data between security tools – sharply reducing manual work and saving around £2.2 million in operating costs.

What they share: start with **one small process that takes a lot of time or effort**, measure the results, then scale. See more at n8n.io/case-studies.

YOUR FIRST WORKFLOW

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Build your first workflow from scratch – in just 5 steps.

- 1 Create an account**

The fastest way: sign up for an n8n Cloud trial at n8n.io. If you'd rather manage your own infrastructure, you can self-host with a single Docker command.
- 2 Pick a repetitive task you want to automate**

Start small: "save invoice attachments to Drive", "post deadline reminders to the team chat every morning". Small wins show results fast.
- 3 Add a trigger, then add action nodes**

Choose the right trigger (schedule, webhook, app event), connect accounts via credentials, then add processing nodes after it. Or get started faster with a template.
- 4 Test step by step**

Click "Execute step" to see each node's input/output data. Adjust until the result is exactly what you want – all in a safe test environment.
- 5 Activate the workflow – and let n8n run it**

Flip the Active switch and the workflow runs automatically whenever its trigger fires. Watch the Executions tab for the first few days to make sure everything runs smoothly.

GOLDEN RULE

Don't try to automate an entire complex process from day one. **One small workflow that works** is worth more than many complex workflows that never get used.

CHOOSE HOW TO DEPLOY

Cloud or self-host? Both give you the full n8n experience, but suit different needs.

EASIEST TO START

n8n Cloud

A service run by the n8n team. Sign up and start right away — no installation, no maintenance, automatic updates.

- + Up and running in minutes, with a free trial
- + No server knowledge needed, automatic backups
- + Paid monthly plans, based on workflow executions
- Data runs through n8n's infrastructure (EU)

Best for: individuals and teams who want to focus on processes, not infrastructure.

FULL CONTROL OVER INFRASTRUCTURE

Self-host

Deploy n8n on your own infrastructure (Docker, VPS, on-premise). The Community edition is free under the fair-code license.

- + Data is processed on infrastructure you manage
- + Free, with unlimited workflows and executions
- + Deep customization; Enterprise plan for larger needs (SSO, permissions...)
- You're responsible for installation, updates, backups and security

Best for: technically capable teams with strict data requirements (finance, healthcare...).

Tip for beginners: learn and experiment on Cloud first; once your processes are clear and data-control needs grow, move to self-host — workflows export/import easily as JSON. Latest pricing: n8n.io/pricing.

BEST PRACTICES & COMMON MISTAKES

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A few principles to keep workflows stable and easy to maintain.

DO

- ✓ **Use clear names**
Name workflows and nodes after what they actually do — so they're still easy to understand later.
- ✓ **Set up an error workflow**
A separate workflow that alerts you on errors — so you catch problems early instead of noticing only after a process has stopped.
- ✓ **Test with realistic data**
Test with data as close to real as possible before switching to Active.
- ✓ **Split large processes into sub-workflows**
One workflow, one job — easier to read, fix and reuse.

AVOID

- ✗ **One giant workflow that does everything**
The longer it gets, the harder it is to debug. Split it up by task.
- ✗ **Pasting API keys directly into nodes**
Always use credentials — encrypted, and changing one place updates every workflow.
- ✗ **Forgetting to check the Executions tab**
No error message doesn't mean the workflow is working — check the execution history regularly.
- ✗ **Automating an unstable process**
If a process still changes every week, standardize it first, then automate.

UNDERSTANDING DATA IN n8n

Understand how data works, and you'll understand what a workflow is doing.

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WHAT DOES AN ITEM LOOK LIKE?

```
{  
  "name": "Trung Tran",  
  "email": "trung@n8nambassador.com",  
  "company": "n8n GmbH"  
}
```

This is an **item** — a data record with 3 **field**:
name, email and company.

Trigger

item

Node

item

Node

Each node **receives data** from the previous node,
processes it and **passes the result** to the next node.

Item

One data record flowing through a workflow — think of it as a row in a spreadsheet.

Field

One piece of data in an item, e.g. `email` — like a cell in a spreadsheet.

JSON

How n8n represents data, as "field name: value" pairs.

DON'T WORRY

Don't know JSON? No problem — you don't need to code to get started. For now, just understand that data is organized into fields like **name**, **email** or **phone**.

EXPRESSION

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Expressions – using data from earlier steps in your workflow.

Example: a sign-up form collects user details, and you want the Gmail node to automatically send a personalized email to each new sign-up.

FORM NODE RECEIVES DATA

```
{  
  "name": "Trung Tran",  
  "email": "trung@n8nambassador.com"  
}
```

IN THE GMAIL NODE

```
To: {{ $json.email }}  
Subject: Hello {{ $json.name }}
```

`$json` – the current item's data

`.email` – gets the value of the email field

`.name` – gets the value of the name field

Form: Trung submits details

Gmail: automatically writes "Hello Trung"

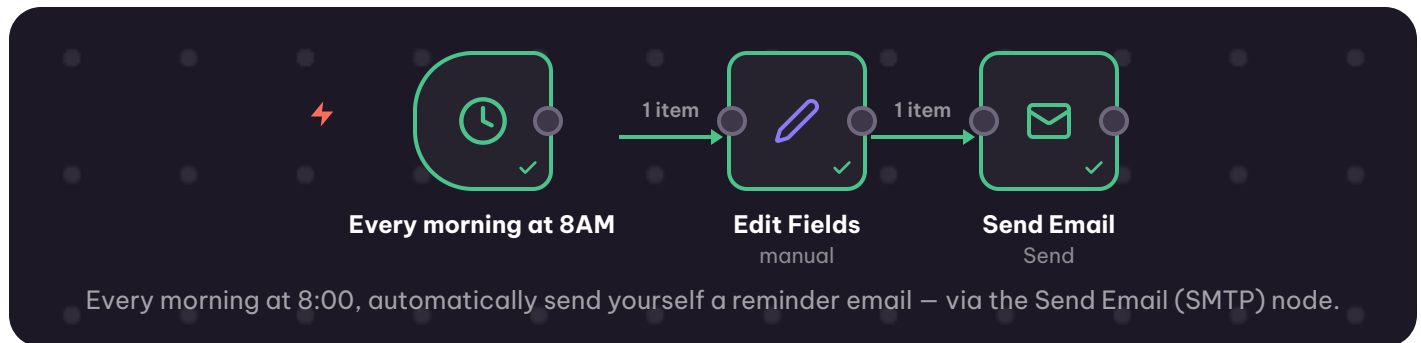
REMEMBER

Put simply, an Expression means: **"Take the data from here and use it over there."** At the beginner level, understanding this principle is enough – you can explore advanced Expressions later.

PRACTICE #1

Build your first workflow in 10 minutes – no AI, no API key needed.

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- 1 Create a new workflow
- 2 Add a **Schedule Trigger**
- 3 Set the run time: 8:00 AM
- 4 Add a **Send Email** node, choose the Send operation
- 5 Create an SMTP credential: host smtp.gmail.com, port 465, SSL/TLS on
- 6 Write the reminder email content
- 7 Click **Execute Workflow** to test it
- 8 Open your inbox and check the email
- 9 Save the workflow
- 10 Activate the workflow – done!

NOTE

For Gmail, use an **App Password** (not your regular password) for the SMTP credential – create one under Google Account → Security → 2-Step Verification. Guide: <https://docs.n8n.io/integrations/builtin/credentials/send-email/gmail>.



Congratulations – you’ve just built your first automated workflow! From here, you can keep adding nodes, data and logic to build more advanced workflows.

IF & BRANCHING

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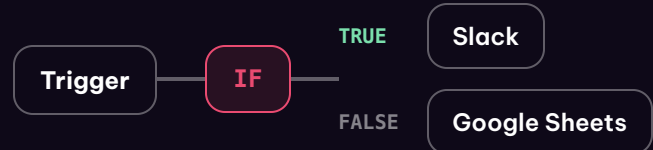
Add logic so your workflow knows what to do in each case.

BASIC WORKFLOW



Always performs the same action on every input.

BRANCHING WORKFLOW



If `priority = high` → notify Slack. Otherwise → save to Google Sheets.

Condition

The condition to check – e.g. "is priority equal to high?"

True

When the condition is true – data follows the True branch.

False

When the condition is false – data follows the False branch.

PICTURE IT

Think of IF as an "If... then..." statement – letting your workflow take different actions depending on conditions, without writing code.

TRY IT

Create a condition: if the order value is > \$1,000 → send a notification to Slack.

DEBUG

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Workflow not behaving as expected? Check each step in order, from left to right.

- 1 Did the trigger fire?**
Click Execute Workflow or create a new event, such as submitting a form or sending a test email.
- 2 Did the previous node return data?**
Click the node and check its OUTPUT. If there's no data, check the nodes before it.
- 3 Does the data have the field you need?**
You used `{{ $json.email }}` but the actual field is named `emailAddress` ? Check the real field name in OUTPUT.
- 4 Is the credential still valid?**
The token may have expired – check authentication and reconnect if needed.
- 5 Check the Executions tab**
Open the failed execution, find the red node and read the error message to identify the cause.



When debugging, check your workflow from left to right. Find the first node whose OUTPUT isn't what you expected – that's usually where the problem starts.

LEARNING RESOURCES

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Finished this guide? Here's what to learn next.

 Official docsdocs.n8n.io

Step-by-step guides, plus beginner courses (Level 1 & 2) with certificates.

 Template libraryn8n.io/workflows

Thousands of workflows by topic: marketing, AI, operations, data... Import and customize to your needs.

 Communitycommunity.n8n.io

The official forum – Q&A, workflow sharing and new feature updates.

 Video tutorialsyoutube.com/@n8n-io

The official YouTube channel: beginner tutorials, AI Agent demos and practical tips.

QUICK GLOSSARY

Canvas	The visual space for building workflows in the editor	Webhook	A URL that receives signals from other systems to start a workflow
Expression	An expression in the form <code>{{...}}</code> used to fetch or transform dynamic data	Item	One data record (JSON) flowing through a workflow
Fair-code	A licensing model that lets you view and use the source code, with some limits on commercial use	Sub-workflow	A workflow called by another workflow to reuse part of a process



Which repetitive task is eating up most of your time this week?

Pick one task. Build your first workflow. Then let n8n handle the repetitive work.

n8n.io

docs.n8n.io



CONNECT WITH ME

Scan the QR code to connect on LinkedIn – I'm an **n8n ambassador**, happy to chat and support the community on its automation journey.